

Question ID: fcdf87b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$y = x^2 - 4x + 4$
 $y = 4 - x$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

Rationale

The correct answer is either 0 or 3. For an ordered pair to satisfy a system of equations, both the x - and y -values of the ordered pair must satisfy each equation in the system. Both expressions on the right-hand side of the given equations are equal to y , therefore it follows that both expressions on the right-hand side of the equations are equal to each other: $x^2 - 4x + 4 = 4 - x$. This equation can be rewritten as $x^2 - 3x = 0$, and then through factoring, the equation becomes $x(x - 3) = 0$. Because the product of the two factors is equal to 0, it can be concluded that either $x = 0$ or $x - 3 = 0$, or rather, $x = 0$ or $x = 3$. Note that 0 and 3 are examples of ways to enter a correct answer.